

GROUNDWATER MODELLING

Can we understand what controls groundwater systems in the Murray Darling Basin (Lower Namoi) through simple analysis?

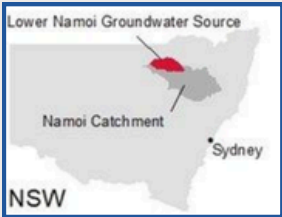
Background

Objective

Lower Namoi groundwater is under pressure from climate variability and extraction, this project asks:

- which climactic fators most strongly control groundwater?
- can simple data-driven water balance model reproduce storage change to predict future scenarios?

Chosen bore GW030450 is a shallow unconfined monitoring bore with 50 years of record



Data and Methods

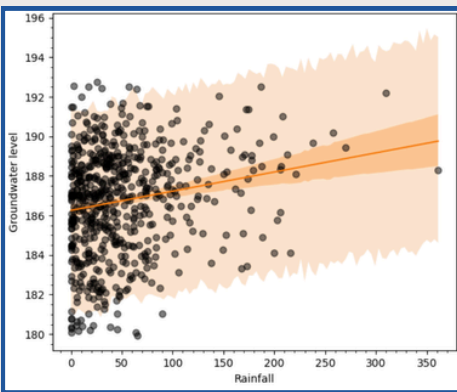
Raw Data → Montly resampling →
Frequency analysis → Cross correlation → Multiple regression →
Water balance model → Scenario projections

Data Sources / References

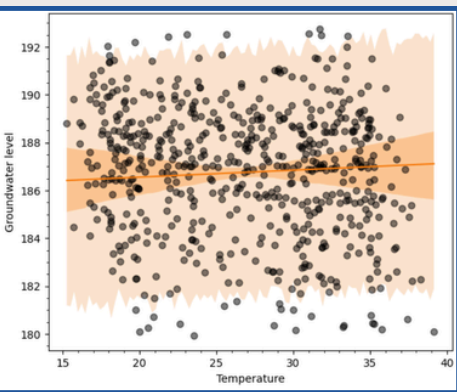
BOM NGIS (Groundwater Explorer)
SILO climate (grid at -30.26, 149.59)
Lower Namoi hydrogeological conceptual model report (DCCEEW)

Key Results: Regression

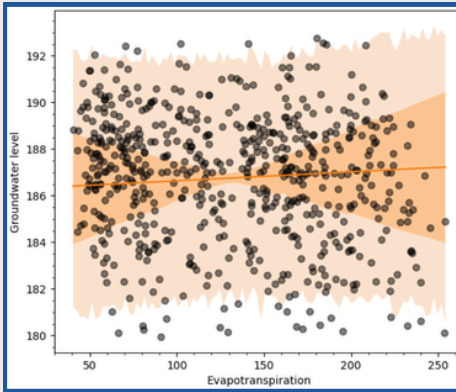
Rainfall: +0.49



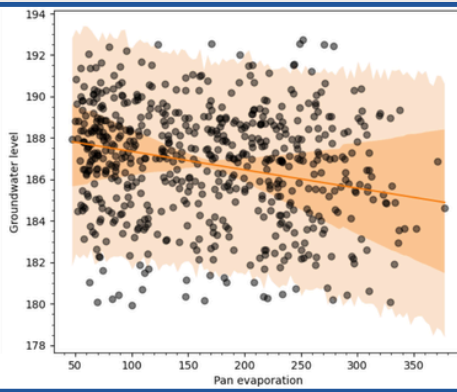
Temperature: +0.17



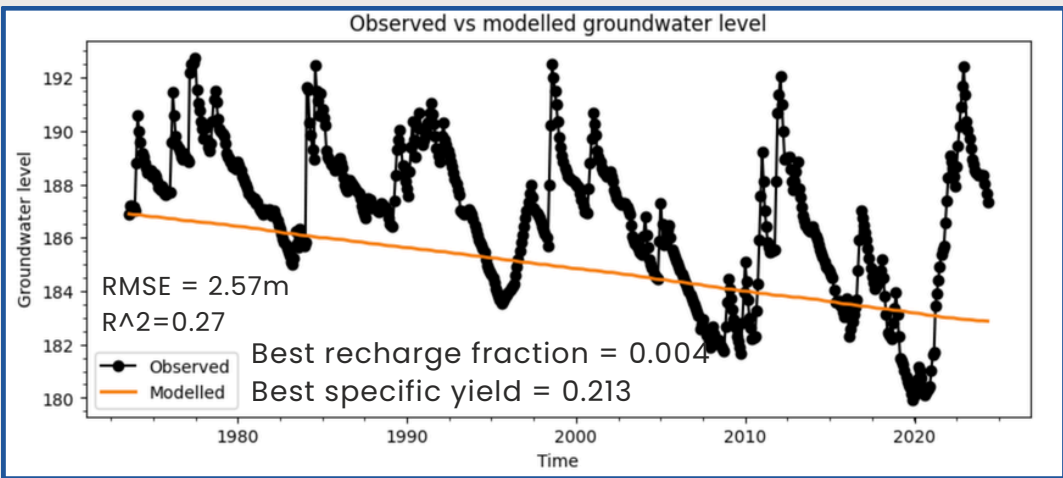
Evapotranspiration: +0.21



Evaporation: -0.71



Model and Scenarios



Millenium Drought
Climate Anomalies:

Rainfall	ET	Evap
-6.20%	3.60%	8.00%

